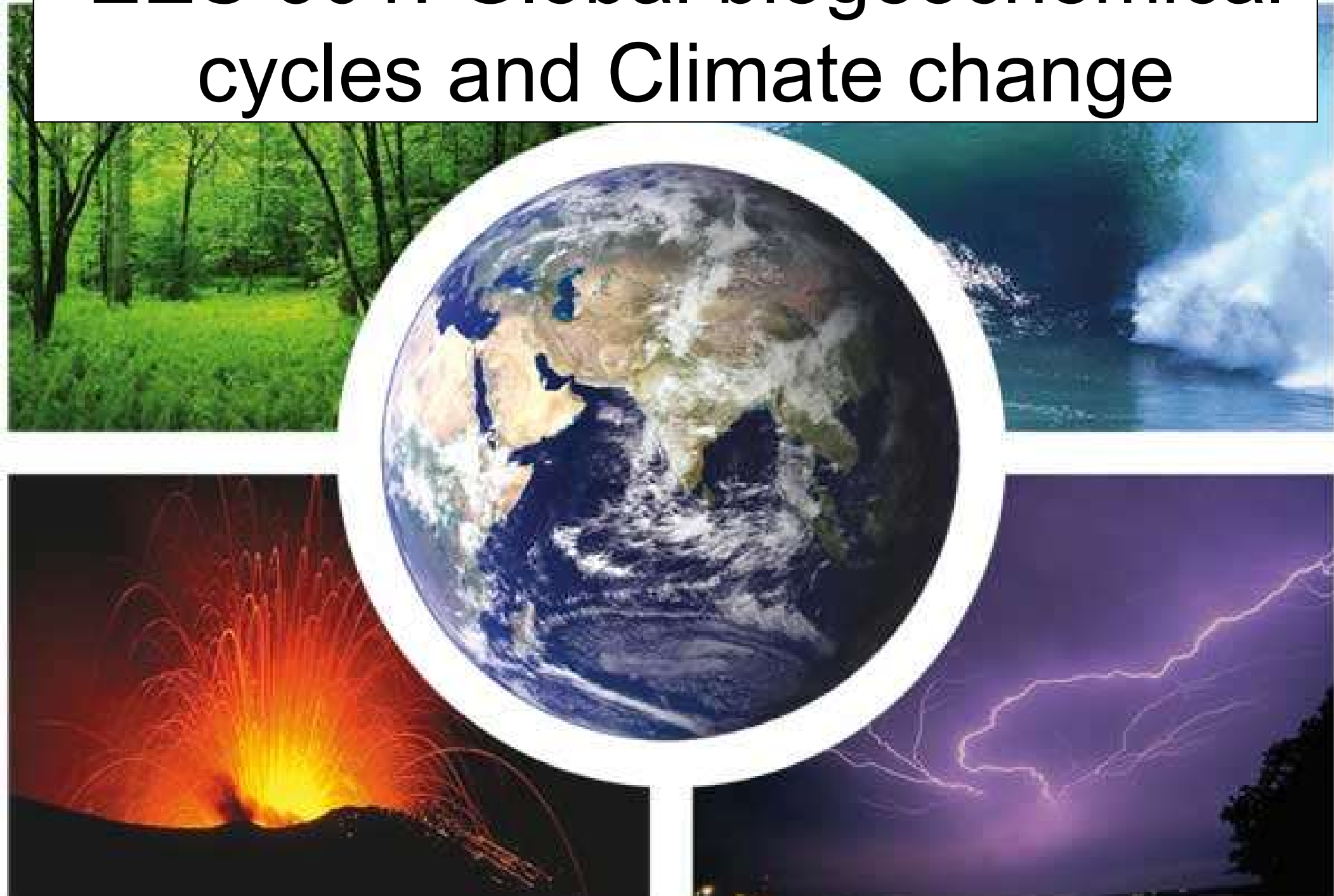


EES 301: Global biogeochemical cycles and Climate change



Reservoir turnover times

Oceans	1.338×10^9	2640 yrs
Cryosphere	24.1×10^6	8900 yrs
Groundwater/permafrost	23.7×10^6	515 yrs
Lakes/rivers	189 990	4.3 yrs
Soil moisture	16 500	52 days
Atmosphere	12 900	8.2 days
Biomass	1120	5.6 days

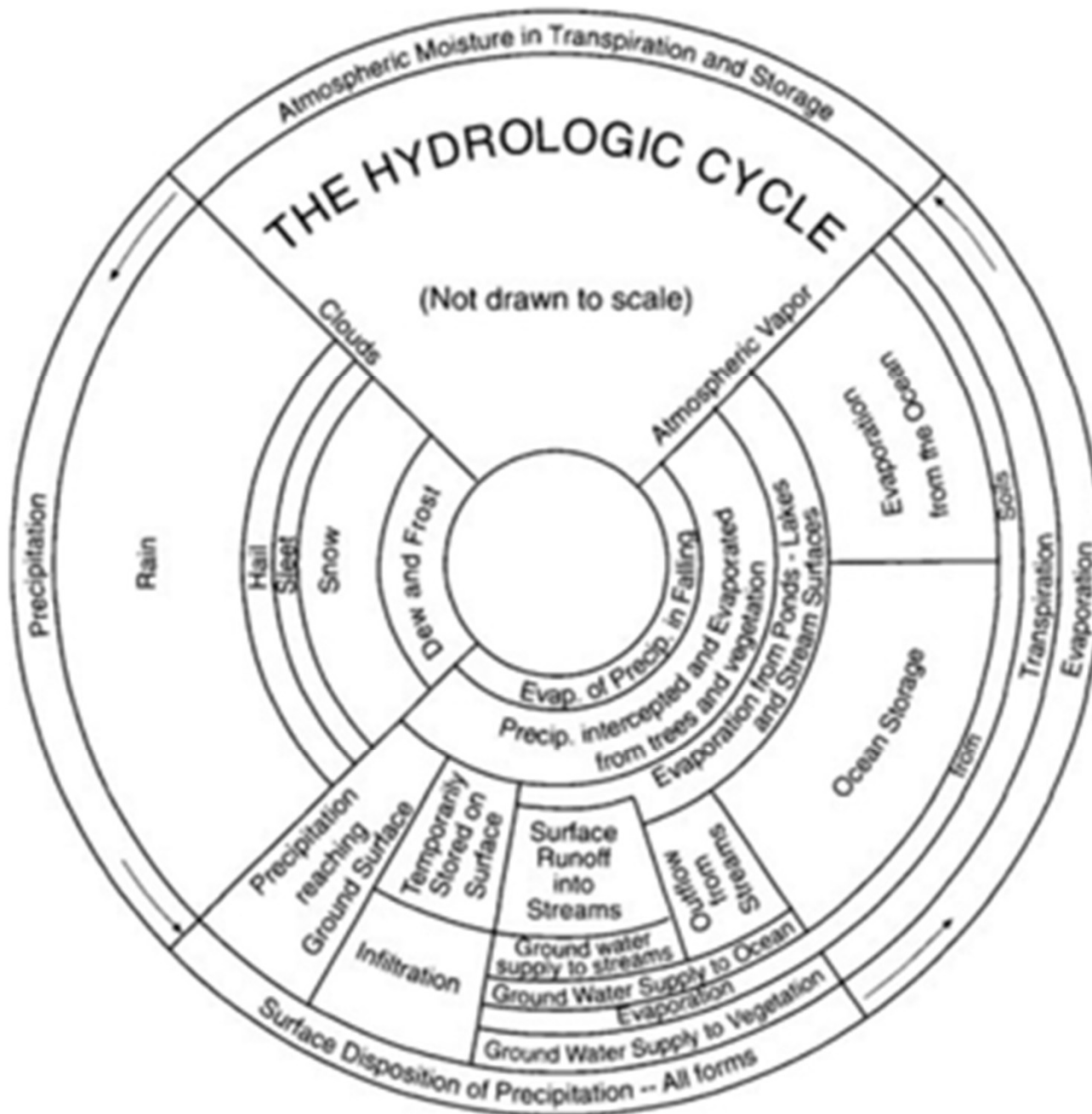
Global average turnover times are used in most books but certain parts of each reservoir can have very different turnover times (e.g. water resides in the Pacific longer than the Atlantic, water in confined aquifers may have turnover times on the scales of thousands or millions of years while the turnover time of unconfined aquifer may be on the scales of months to years)

Hillslope scale turnover times

Reservoir	Depth (mm)	Turnover times
Atmosphere	25 ^a	8–10 days
Plants	5–50	Hours-days
Streams/rivers	3	Weeks
Lakes/reservoirs	—	Months-years
Soil moisture	10–10 ⁴	Years
Groundwater	10 ⁴ –10 ⁵	Days-10 ⁶ years ^b

Most rain falls over mountains where sediment layers are thin and unstable and gravity moves water quickly. Depths are estimated as typical volume averaged over a mountainous watershed areas

Fluxes in the hydrological cycle



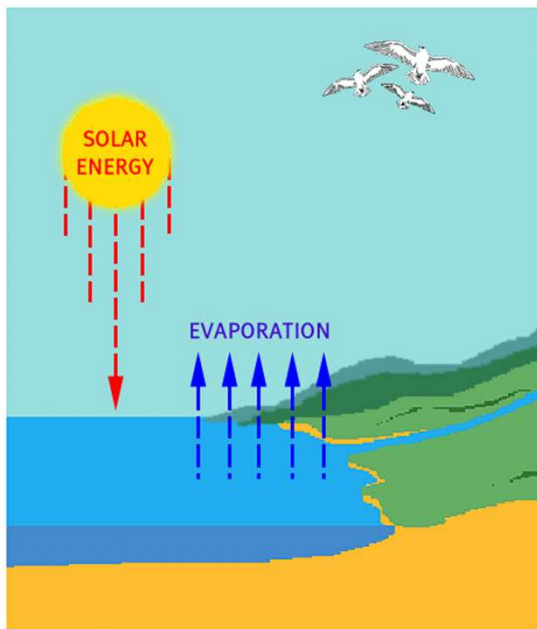
The Horton wheel by Robert Horton

Evaporation and transpiration transfer water into the atmosphere

Evaporation: Conversion of water from a liquid into a gas. Water is transferred from surface waters and oceans to the atmosphere through evaporation **90%**

Transpiration: The process of water loss from plants (plants sweating) through stomata. Water is transferred from soil moisture to the atmosphere through transpiration **10%**

Evaporation



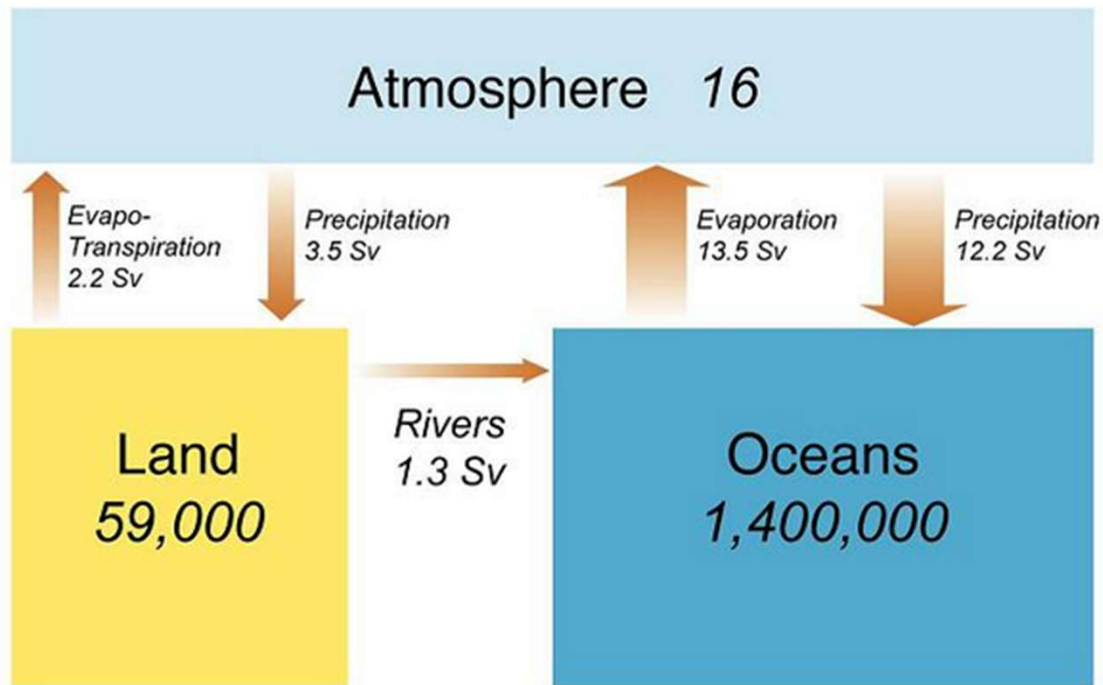
- As water is heated by the sun, its surface molecules become sufficiently energized to break free of the attractive force binding them together, and then evaporate and rise as invisible vapor in the atmosphere.

- 80% of evaporation from ocean most falls back into the ocean as rain

- 20% of evapotranspiration from inland water and vegetation

- Winds transport evaporated water, influencing humidity around the world

Global Water Reservoirs and Fluxes



Reservoirs in 10³ km³, Fluxes in 10⁶ m³/s (= Sv)

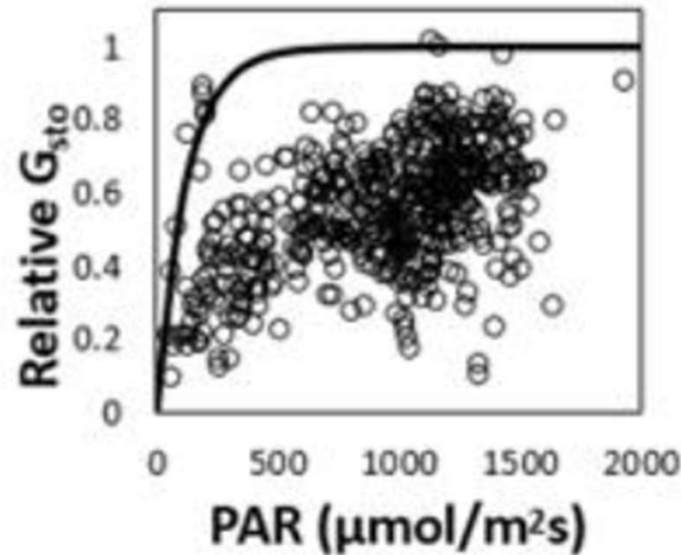
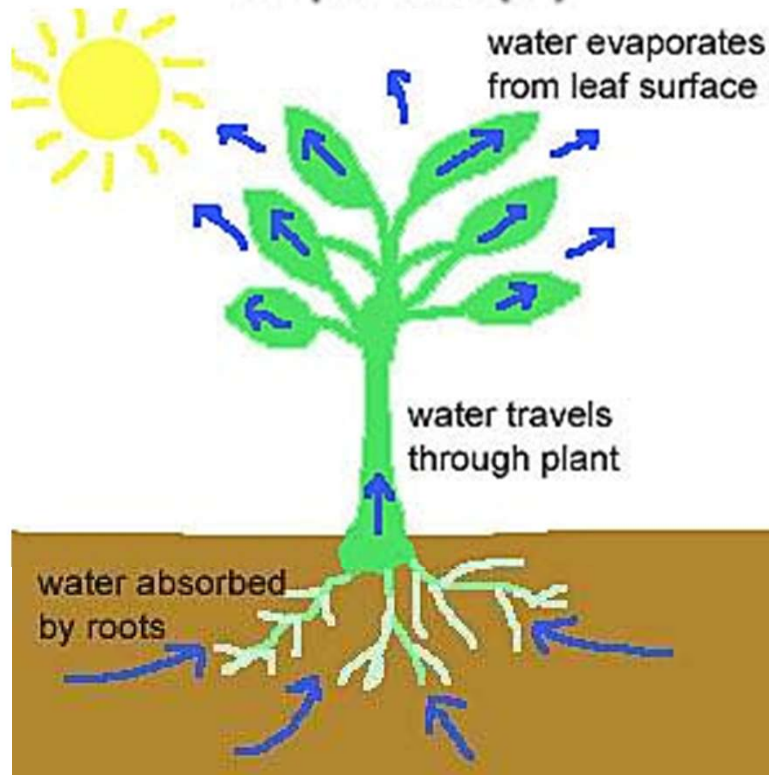
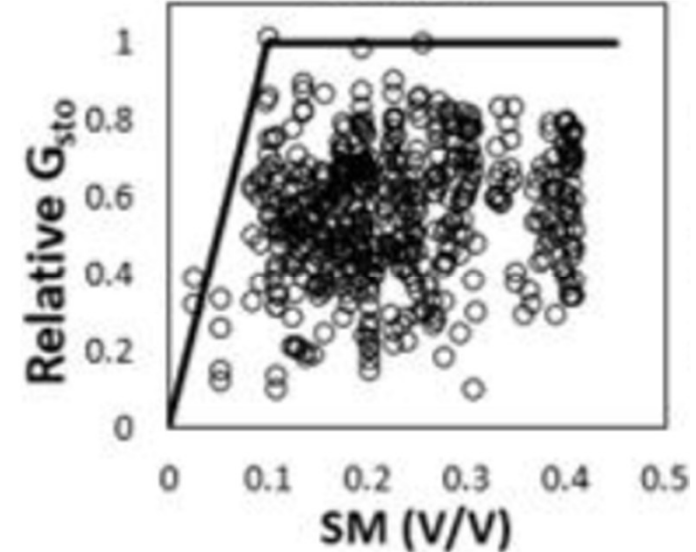
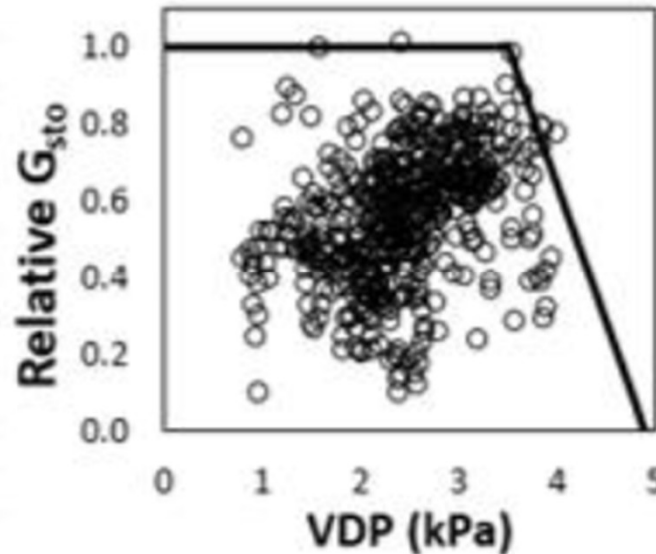
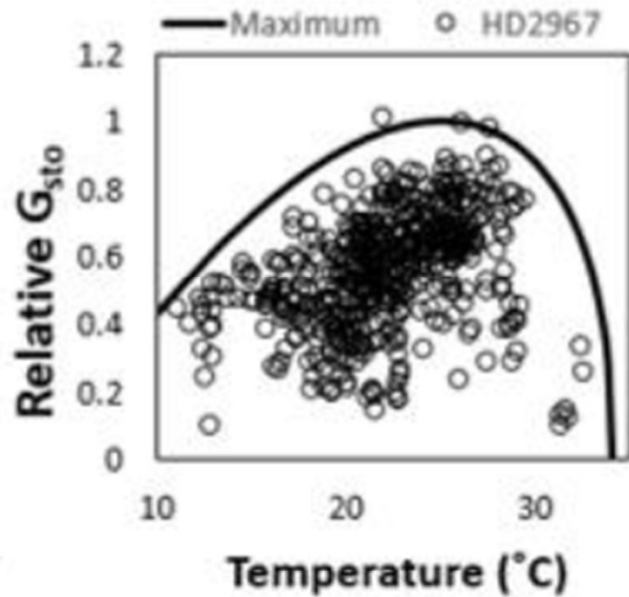
Transpiration

The process of water loss from plants (plants sweating) through stomata.

(Stomata are small openings found on the underside of leaves that are connected to vascular plant tissues.)

- passive process that depends on:
 - ~humidity of the atmosphere
 - ~the moisture content of the soil
 - ~How the air temperature relates to the temperature optimum for photosynthesis
- only 1 % of the water transpired used for growth
- transports nutrients from the soil into the roots and carries them to the various cells of the plant
- keeps tissues from becoming overheated

Transpiration



Environmental limitations determine the net water transfer

The behavior of different species can be measured

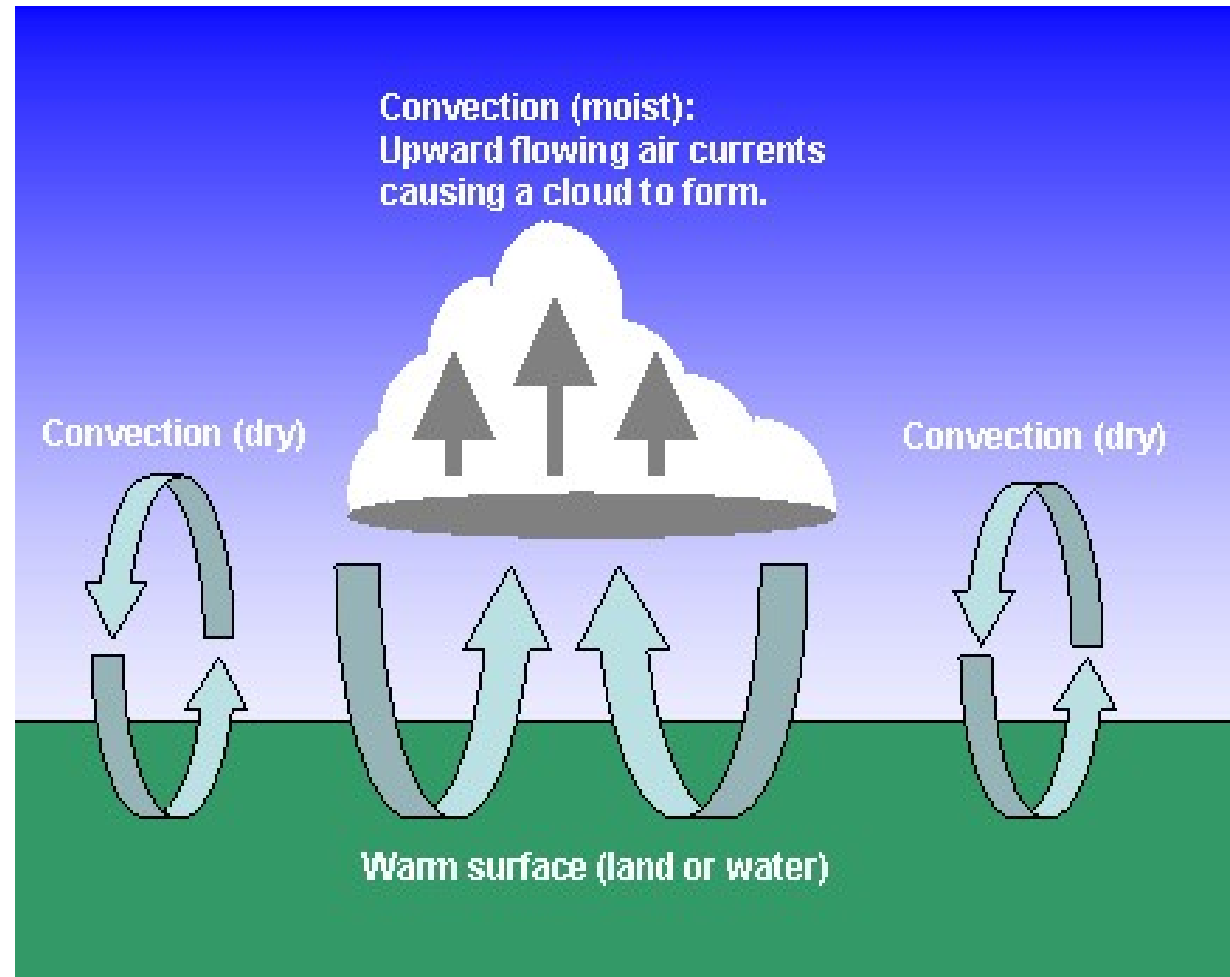
Every day an actively growing plant transpires 5 to 10 times as much water as it can hold at one time.

Condensation & Precipitation

- Condensation and Precipitation move water from the atmosphere to the land or ocean surface.
- Condensation is the process of water changing from vapor to liquid. Water vapor in the air rises mostly by convection. This means that warm, humid air will rise by virtue of its lower density ($PV=nRT$), while cooler air will flow downward by virtue of its high density.
- As the warmer air rises to higher levels it cools and relative humidity increases. When it becomes supersaturated water condenses onto aerosols to form droplets.
- At high altitudes significantly below zero °C temperatures water may also freeze onto ice nuclei (microbes and dust)

Condensation & Convection

- Warm air rises
- Air cools and can no longer hold vapor
- Process known as **CONVECTION**
- Cooling of water vapor forms **CLOUDS**
- The level at which this happens is known as **lifting condensation level**



Precipitation

- Precipitation is water being released from clouds rain, freezing rain, sleet, snow, or hail
- Precipitation begins after water vapor, which has condensed in the atmosphere, becomes too heavy to remain in atmospheric air currents and falls. The driving force is gravity.
- Under some circumstances precipitation actually evaporates before it reaches the surface.



Rain

- Type of precipitation during warmer weather
- Occurs when clouds drops become heavy and large
- Falls to earth in the form of water and runs off to streams & rivers

Snow



- Snow is a type of precipitation like rain but forms at cooler temperatures
- Usually reaches the ground and stays
- Eventually melts and becomes runoff in streams
- The spring melt is important for agriculture at high altitudes and in temperate regions
- Snow contributes to albedo

Hail

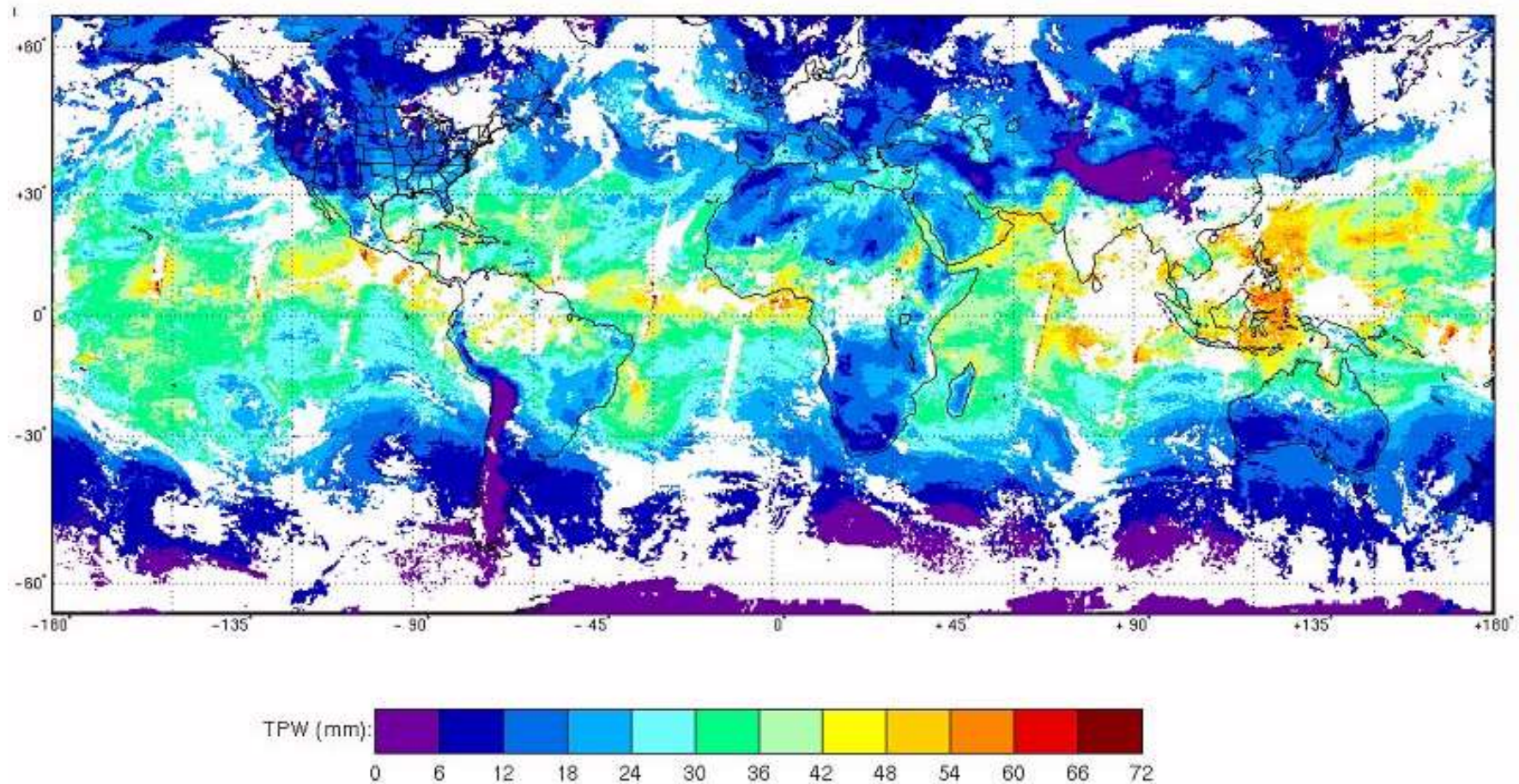


- Hail is a form of precipitation consisting of solid ice that forms inside thunderstorm updrafts.
- Hail can damage aircraft, homes and cars, and can be deadly to livestock and people.

Evaporation vs Precipitation

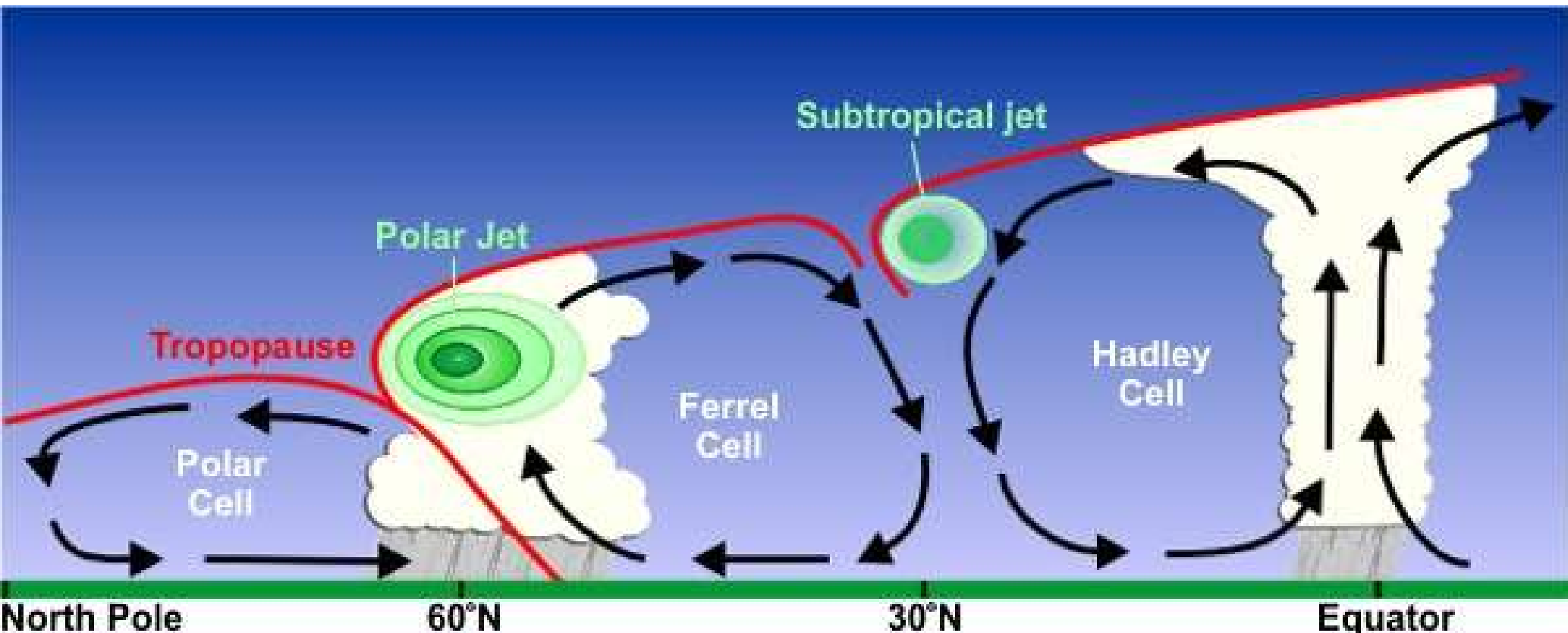
- Both processes equal out on a global scale
- Evaporation more prevalent over the oceans than precipitation
- Over land, precipitation exceeds evaporation
- Most water evaporated from the oceans falls back into the ocean as precipitation
- 10% of water evaporated from the ocean is transported over land and falls as precipitation
- Once evaporated, a water molecule spends ~ 10 days airborne

Total precipitable water and gross evaporation is highest over the oceans in the tropics



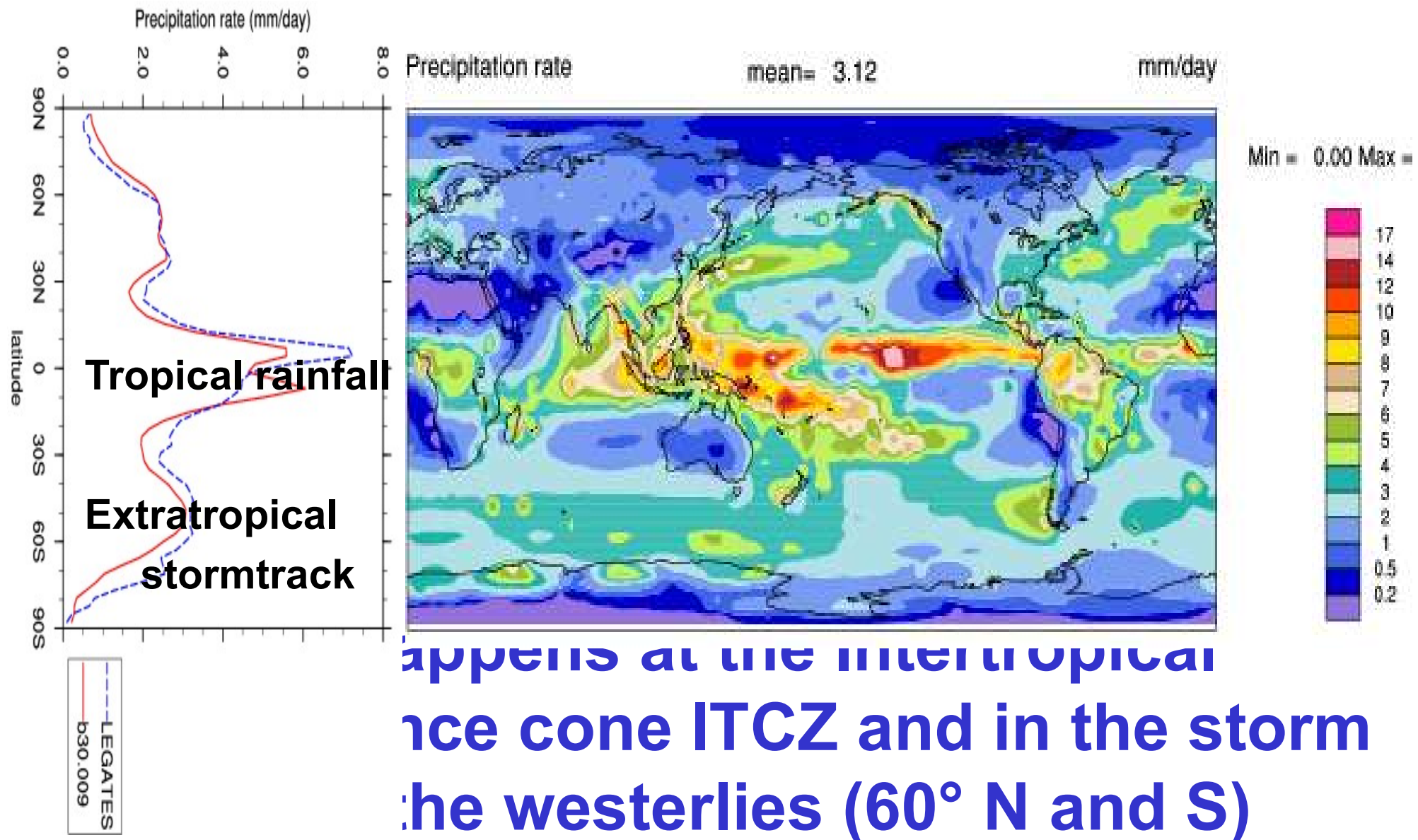
The total atmospheric water vapor contained in a vertical column of unit cross-sectional area from the Earth's surface to the "top of the atmosphere"

Rain is highest where atmospheric cells move the air upwards

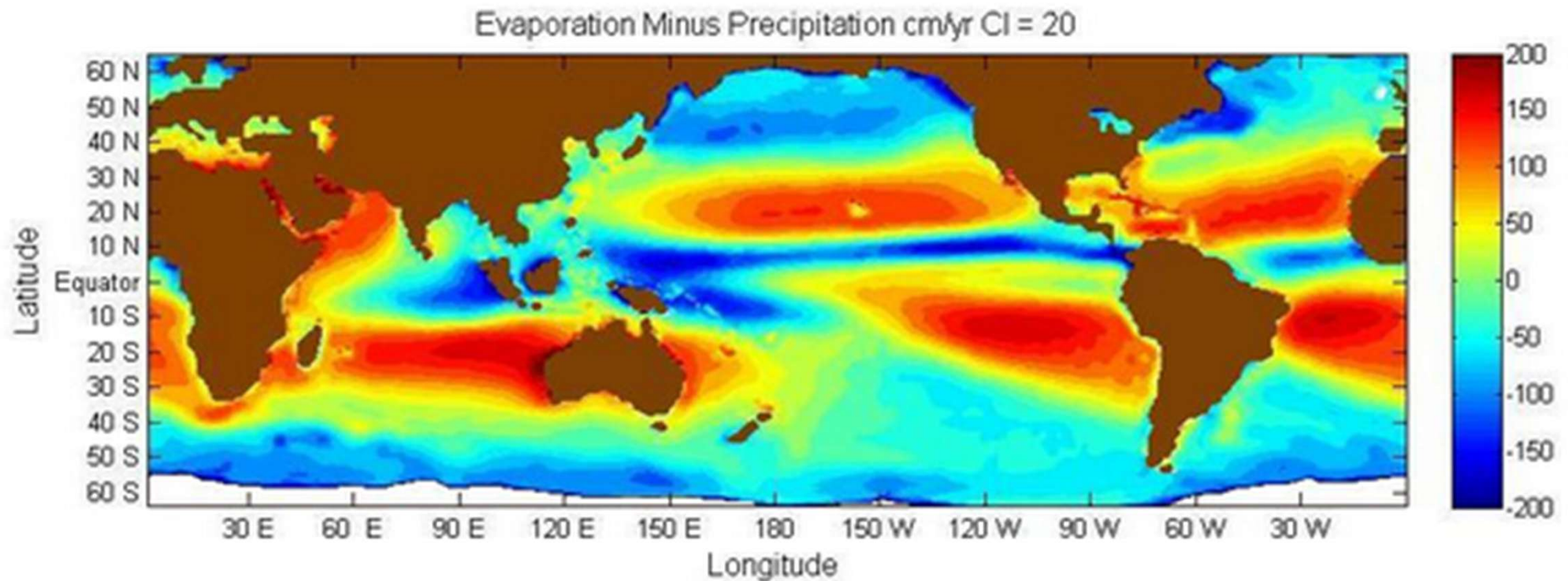


- Rainfall happens at the Intertropical convergence zone ITCZ and in the storm tracks of the westerlies (60° N and S)

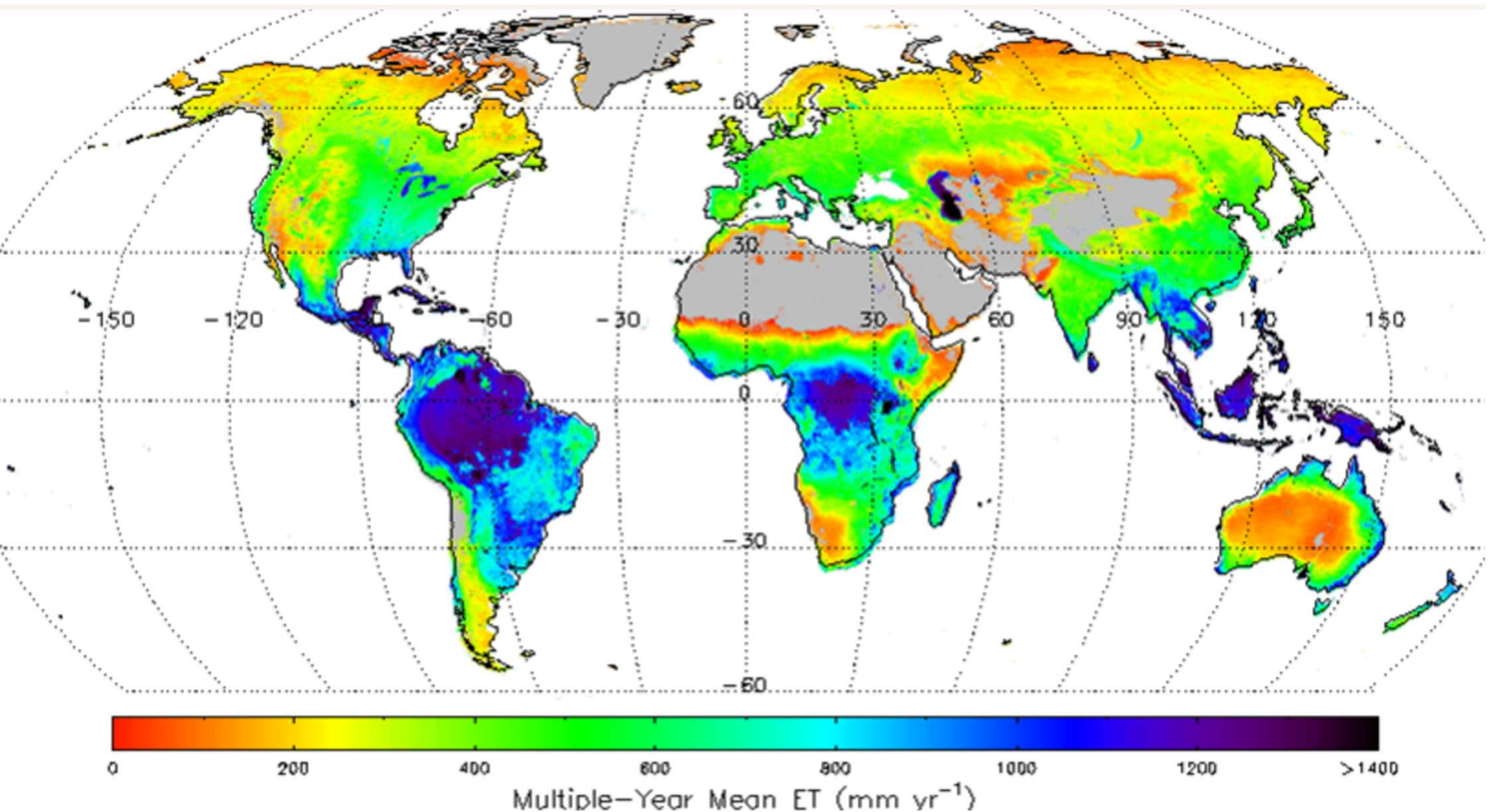
Rain is highest where atmospheric cells move the air upwards



Global net Evaporation Evaporation - Precipitation



Evapotranspiration is highest over ecosystems that receive a lot of rain and have high leaf surface area



Rain does not follow the plough – agricultural crops have low LAI and don't stay year around

**How many liter of water fall in Punjab when
in monsoon approximately 600 mm of rain
fall on 1 ha of land**

$$600 \times 10^{-3} [\text{m}] \times 10000 [\text{m}^2] = \\ 6000 [\text{m}^3] = 6 \times 10^6 [\text{L}]$$

**On average it takes 3000 to 4000 L of water to grow 1
kg of rice and one ha yields 4000 to 7000 kg rice.
How much irrigation is needed to grow rice? What
fraction of the total water is green water (rainfall)
and blue water (irrigation)**

$$3 \times 4 \times 10^6 [\text{L}] \text{ to } 4 \times 7 \times 10^6 [\text{L}] \\ = 12 \times 10^6 [\text{L}] \text{ to } 28 \times 10^6 [\text{L}] \text{ total water} \\ = 6 \times 10^6 [\text{L}] \text{ to } 22 \times 10^6 [\text{L}] \text{ irrigation water} \\ 21\% \text{ to } 50\% \text{ green water}$$

Subsurface Runoff

- **Subsurface runoff starts with infiltration. Infiltration means that a portion of the precipitation that reaches the Earth's surface seeps into the ground.**
- **The amount of water that infiltrates the soil varies with the degree of land slope, the amount and type of vegetation, soil type and rock type, and whether the soil is already saturated by water.**
- **Infiltrated water can reside in the soil as soil moisture or percolate down to groundwater.**
- **Groundwater will move down slope slowly and finally feed into streams**

Wet Period

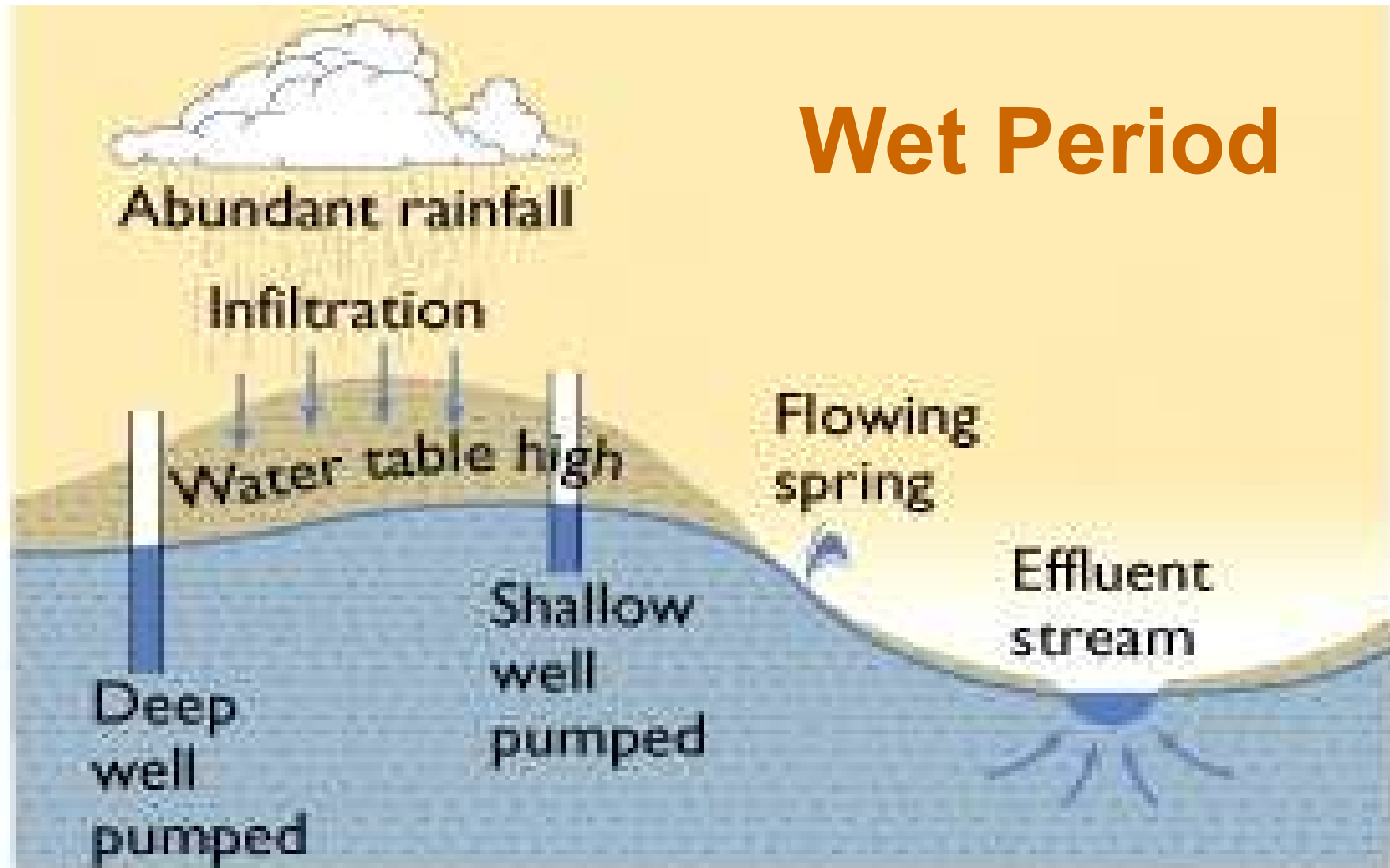


Fig. 12.10

Dry Period

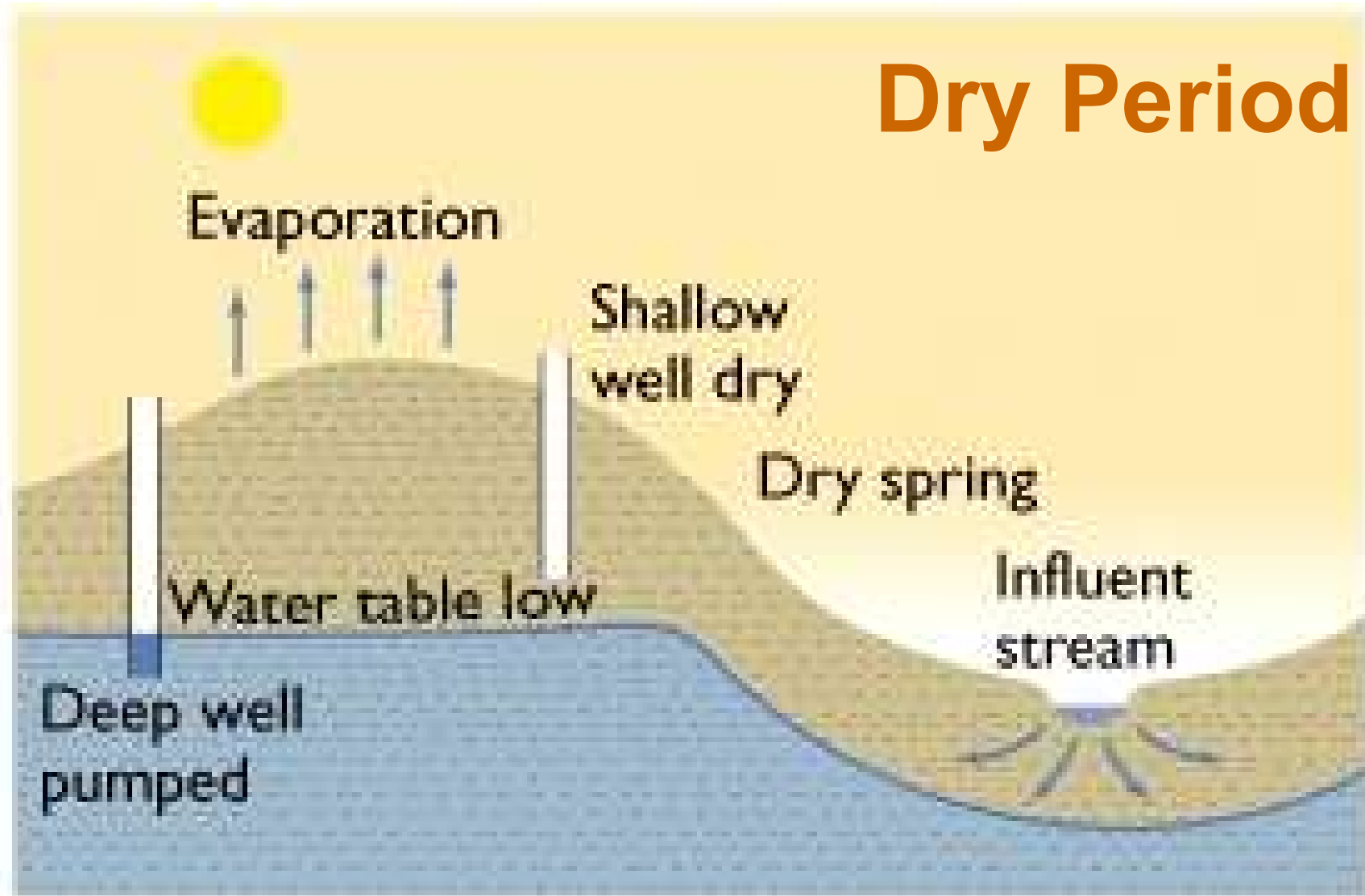


Fig. 12.10

Darcy's Law

The speed of the groundwater flow follows Darcy's Law

$$v = \frac{K [m s^{-1}]}{n} \frac{dh}{dL}$$

K=soil hydraulic conductivity

n= actively available porosity

dh/dL = hydraulic gradient

DARCY'S LAW

Volume of water flowing in a certain time is proportional to:

vertical drop ÷ flow distance
or

$$\frac{Q}{A} = K \times \frac{h}{\ell}$$

Q: Volume of water flowing in a given time

A: Cross-sectional area through which water flows

K: Hydraulic conductivity (a measure of permeability)

h: Vertical drop between two points

ℓ Distance the flow travels

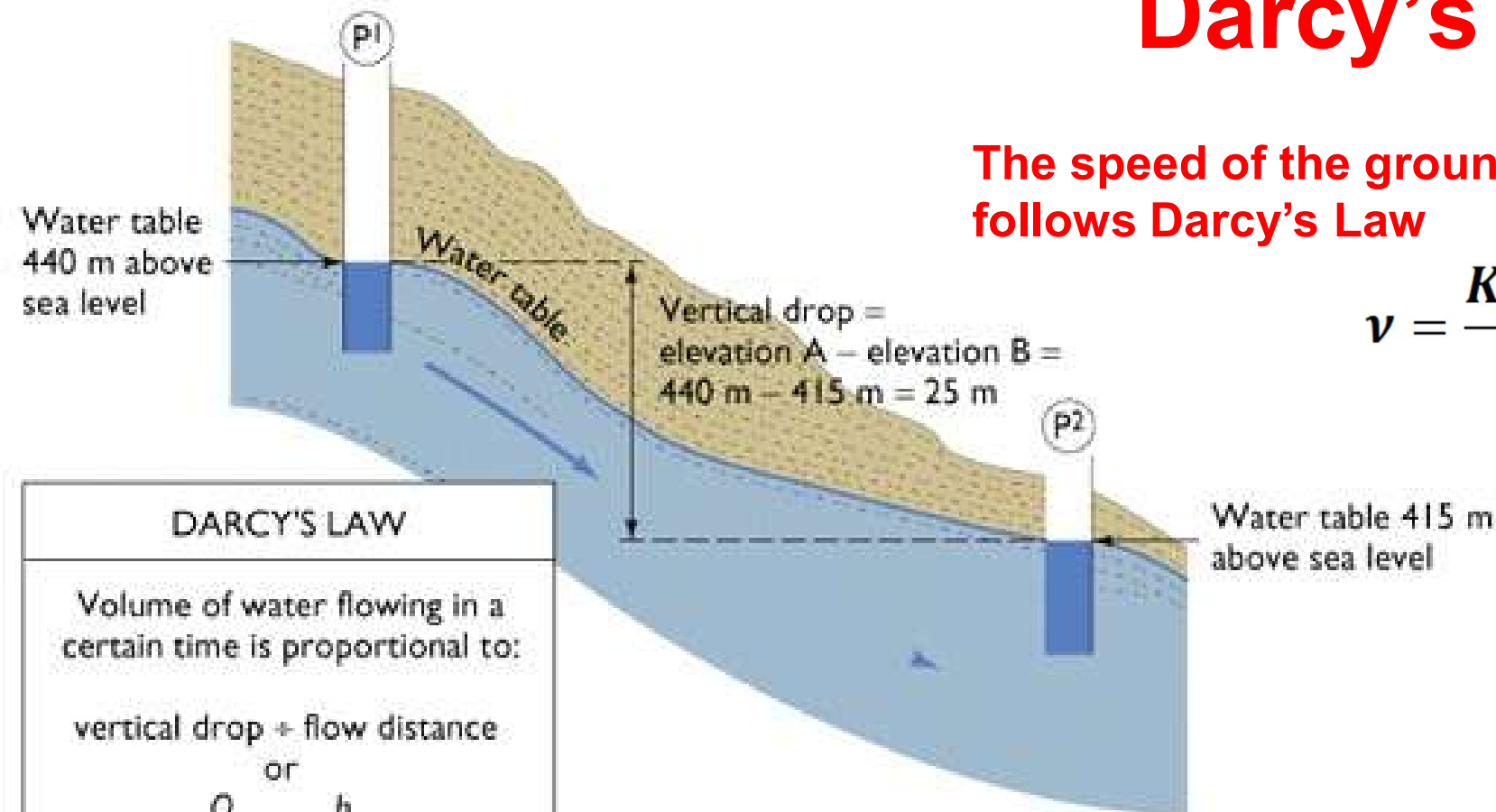


Fig. 12.16

Soils and rocks are *not* completely solid.

porosity: portion of volume of a material that consists of open spaces

permeability: measure of the speed at which fluid can travel through a porous medium

Imagine two vertical pipes, one filled with gravel, one with sand. Out of which one will the water flow faster?

Porous Sandstone

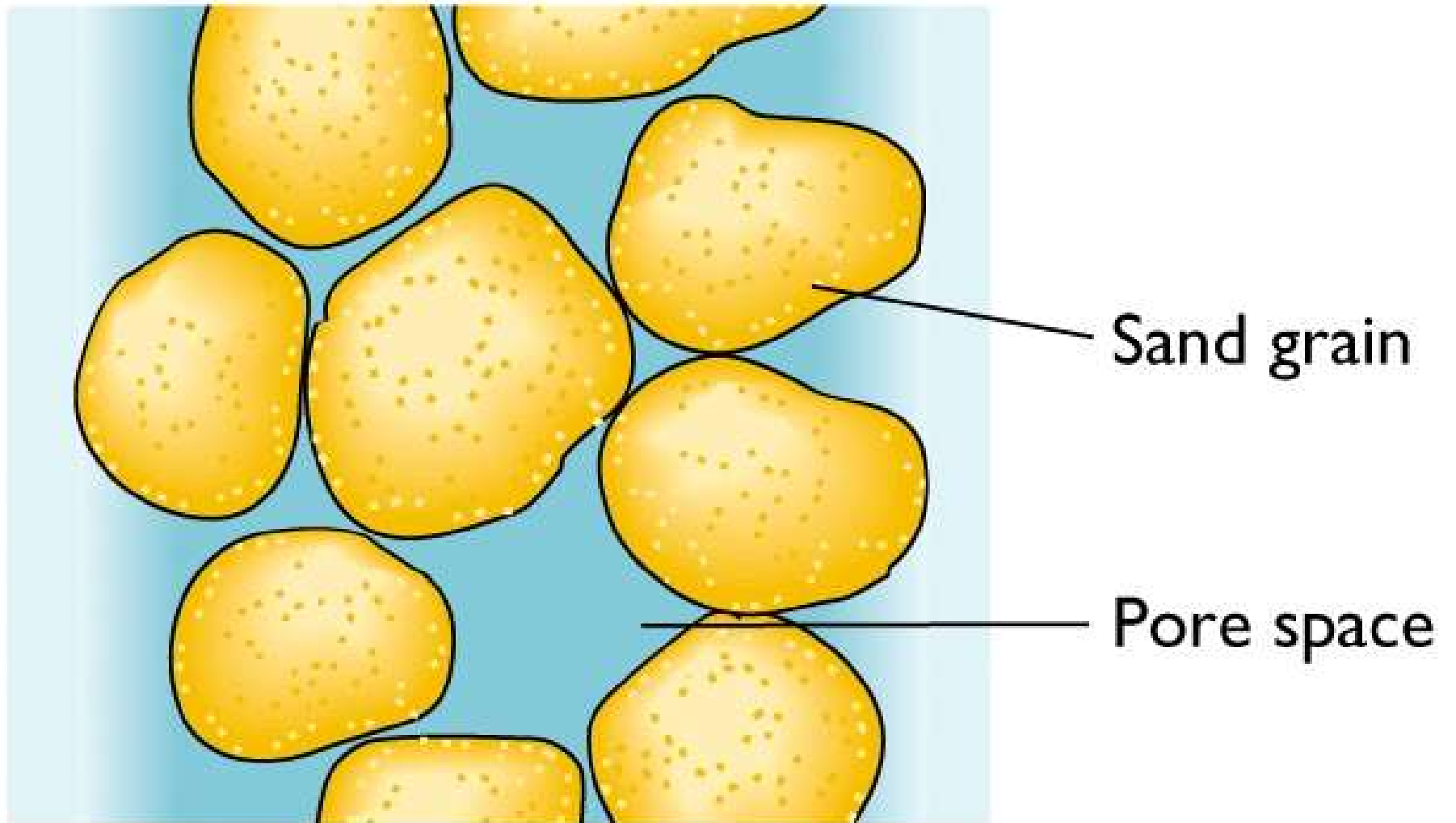


Fig. 12.7a

Cemented Sandstone

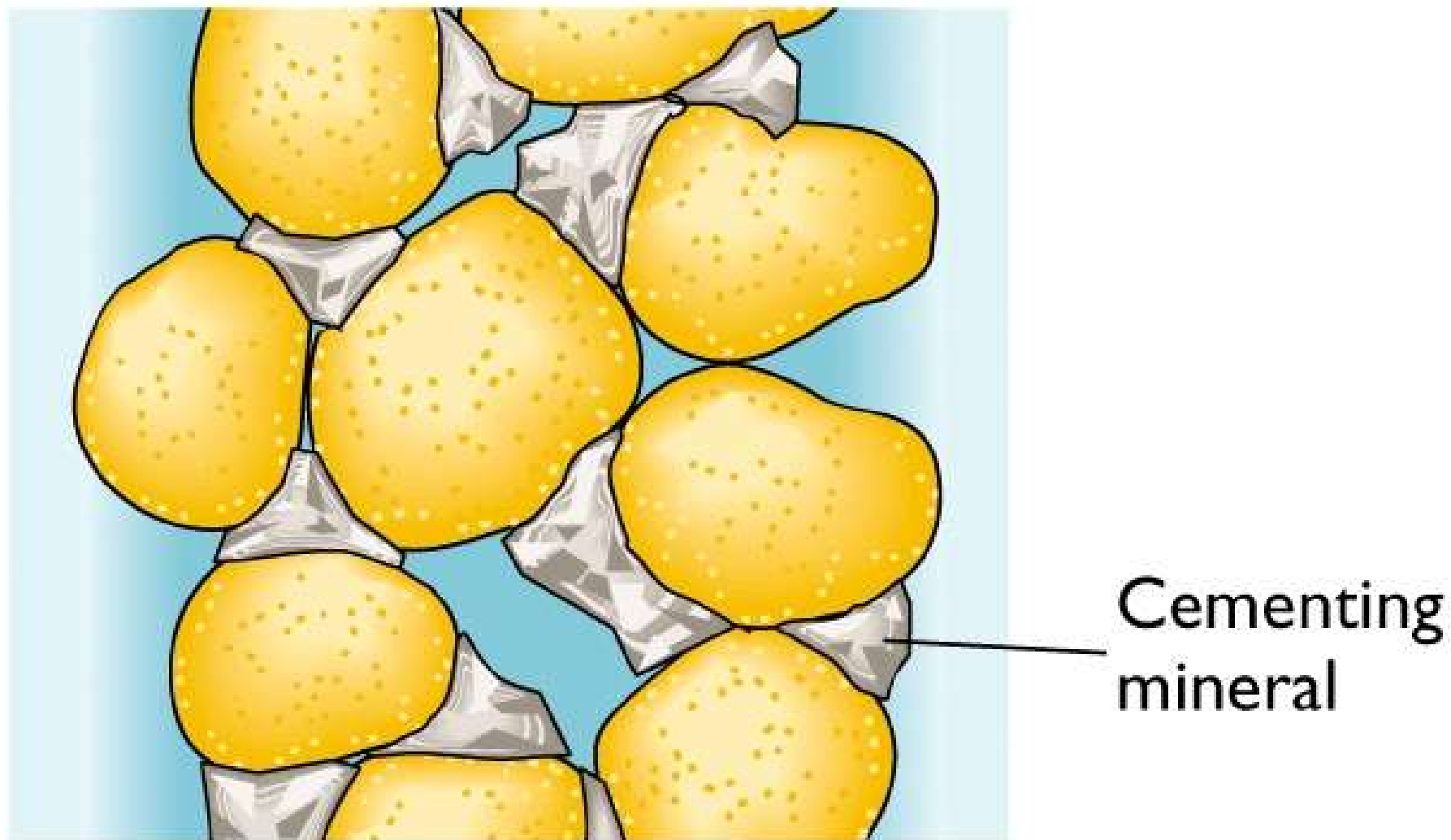
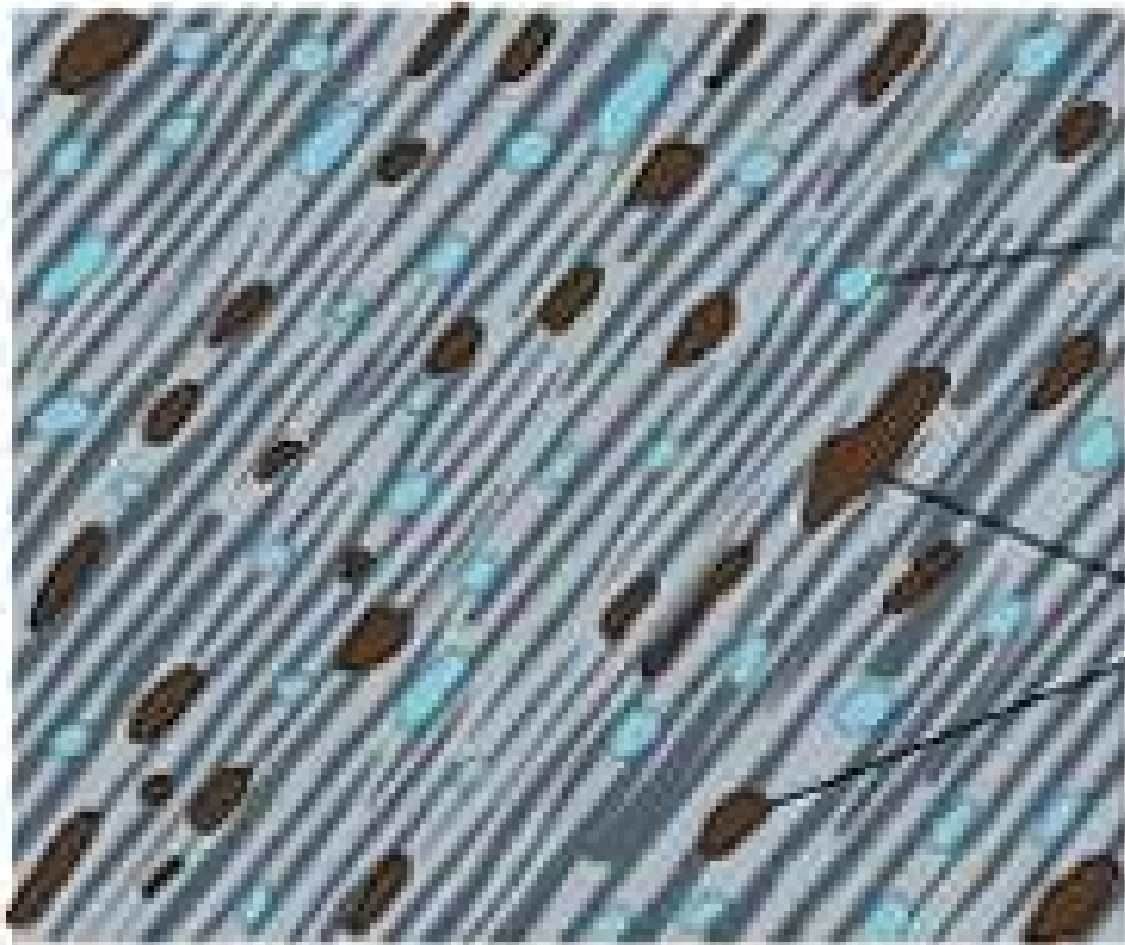


Fig. 12.7b

Unfractured Shale



Very small amounts
of pore space
between clays
and silt grains

Silt grains

Fig. 12.7e

Groundwater terms

aquifer: body of rock that is sufficiently water permeable to yield economically significant quantities to wells and springs

aquitard: body of rock that retards but does not prevent flow of water to or from an adjacent aquifer

aquiclude: body of relatively impermeable rock that is capable of absorbing water slowly but does not transmit it rapidly enough to supply a well or spring

Groundwater Table

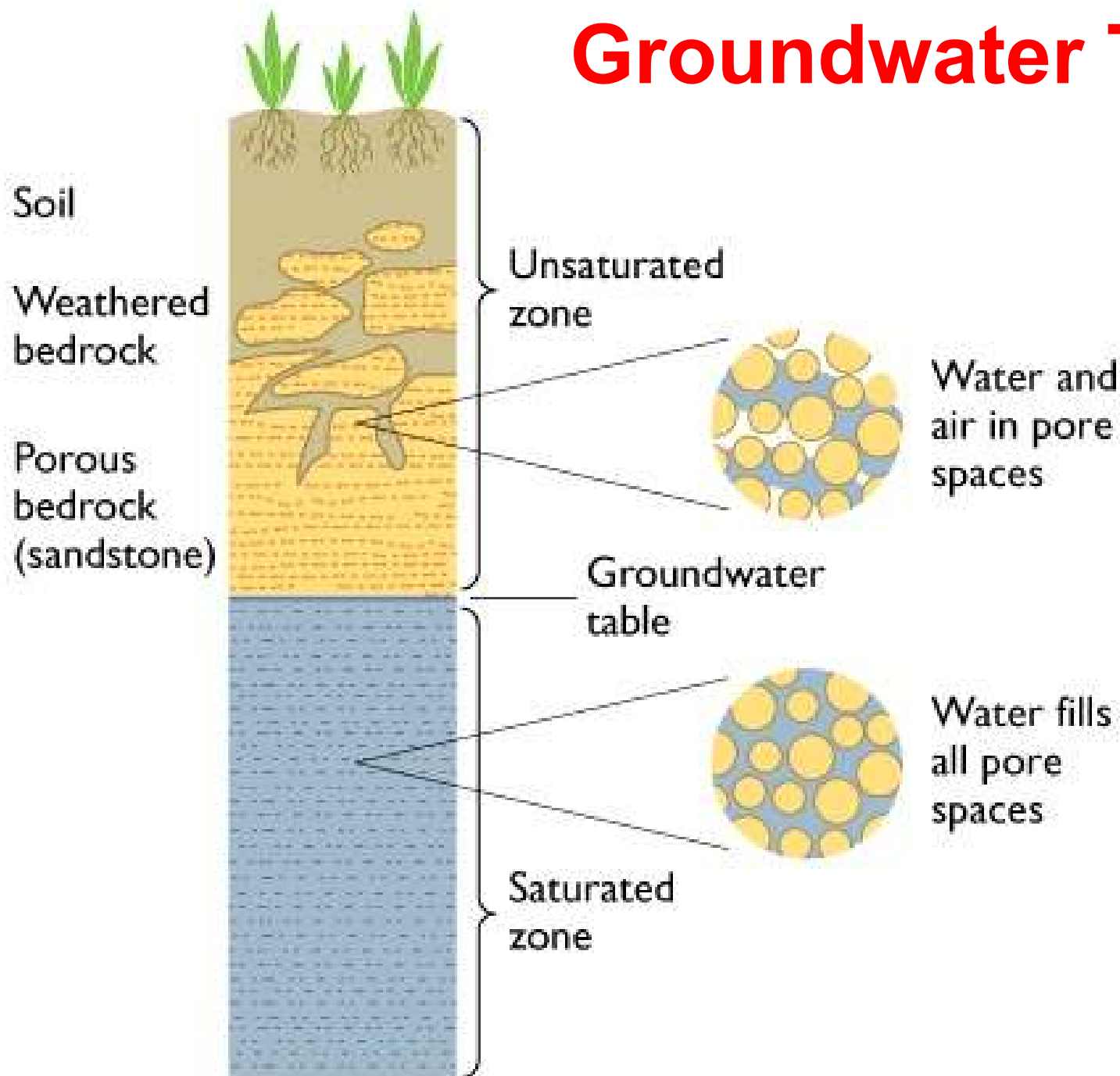
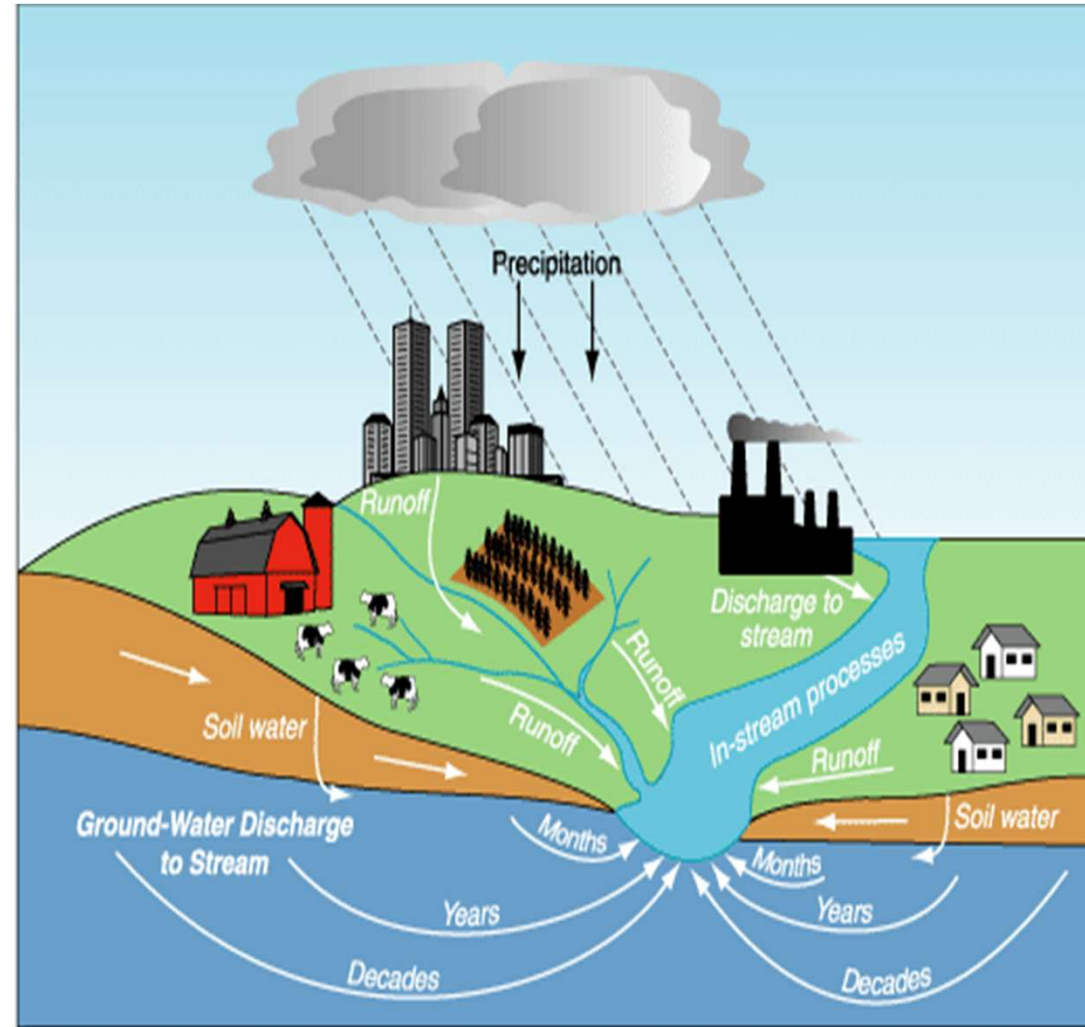


Fig. 12.8

Surface Runoff

Surface Runoff: When there is a lot of precipitation, soils become saturated with water. Additional rainfall can no longer enter it. Runoff will eventually drain into creeks, streams, and rivers, adding a large amount of water to the flow in a short time. Surface water always travels towards the lowest point possible, usually the oceans. Along the way some water evaporates, percolates into the ground, or is used for agricultural, residential, or industrial purposes.



Meteorological factors affecting surface (over soil) runoff

- Type of precipitation**
- Rainfall intensity**
- Rainfall amount**
- Rainfall duration**
- Distribution of rainfall over the drainage basin**
- Direction of storm movement**
- Precipitation that occurred earlier and resulting soil moisture**
- Meteorological conditions that affect evapotranspiration**

Physical characteristics affecting surface runoff



Overland runoff from disturbed areas often contains excessive sediment in addition to water. (USGS)

- Land use
- Vegetation
- Soil type
- Drainage area
- Basin shape
- Elevation
- Topography, especially the slope of the land
- Drainage network patterns
- Ponds, lakes, reservoirs, sinks, etc. in the basin, which prevent or delay runoff from continuing downstream

Types of runoff and impact on streamflow

Saturated overland flow



Horton overland flow

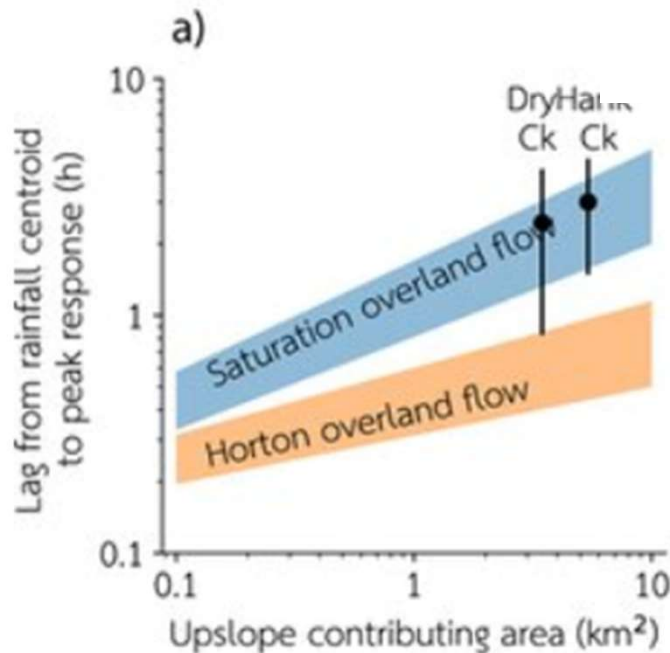
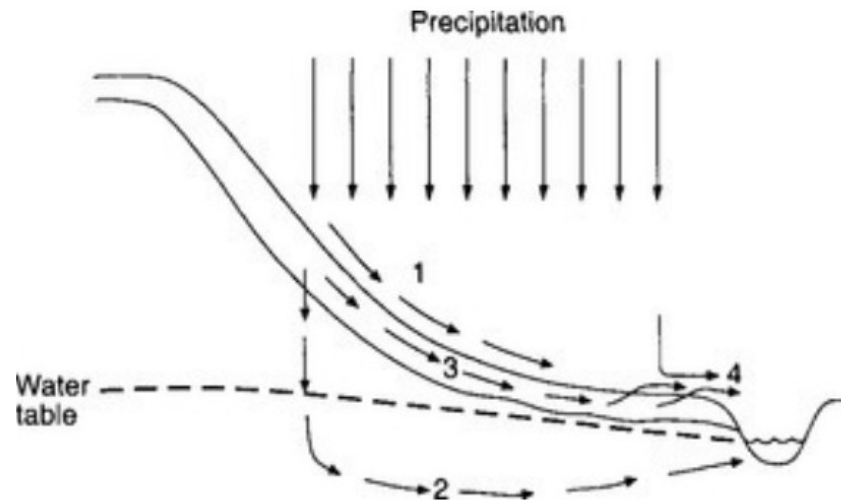


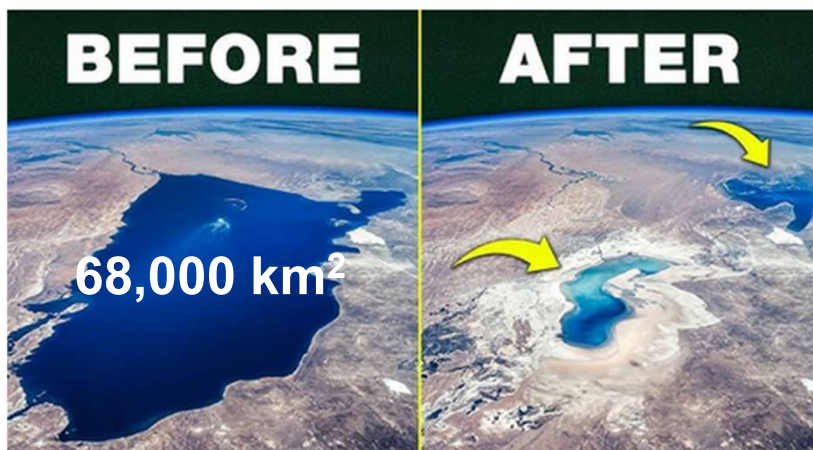
FIG. 6-7 Vertical cross-section showing pathways for surface and subsurface runoff. Path 1: HOF; path 2: groundwater flow; path 3: subsurface stormflow; path 4: SOF. (From [Dunne and Leopold, 1978](#).)

Runoff

- Drains to a creek
 - To a stream
 - To a river
 - To ocean
 - Sometimes runoff drains to a closed lake
 - May be diverted for human uses

World's largest rivers

River	Discharge (m ³ /s)	Discharge(km ³ /yr)
Amazon	190 000	6000
Congo	42 000	1330
Yangtze	35 000	1100
Orinoco	29 000	915
Brahmaputra	20 000	630
La Plata	19 500	615
Yenesei	17 800	565
Mississippi	17 700	560



The hydrological cycle & climate change

- About 3% of precipitation will seep underground to form groundwater
- About 31% will run off into rivers, streams and lakes
- About 66% of the water returns to the atmosphere through evaporation and transpiration
- System responses are nonlinear

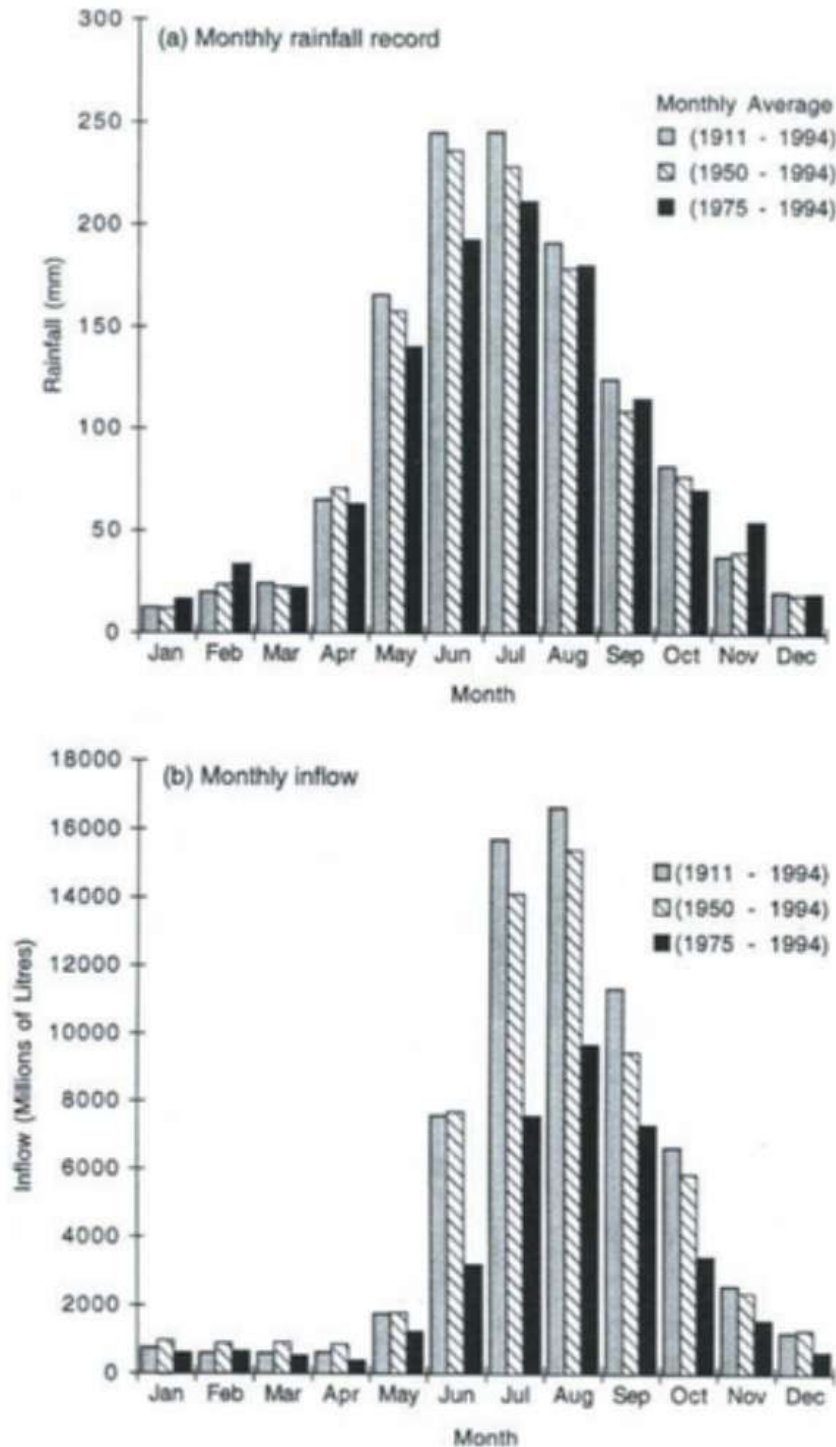


FIG. 6-11 (a) Monthly rainfall record. Serpentine dam, Perth, Western Australia. (b) Monthly inflow. Serpentine Dam, Western Australia. (From [Burges, 1998](#) .)

(From [Burges, 1998](#) .)

Groundwater & Oil have a lot in common

- **Both require an impermeable “trap” layer and a permeable layer with the fluid**
- **Recharge of water is faster than that of Oil which takes millennia to form but ultimately limited by rainfall**
- **Sustainable transport fuels = renewable fuels**
- **Sustainable water management = rainfall limited – other demands have to be met through secondary & tertiary use & recycling**

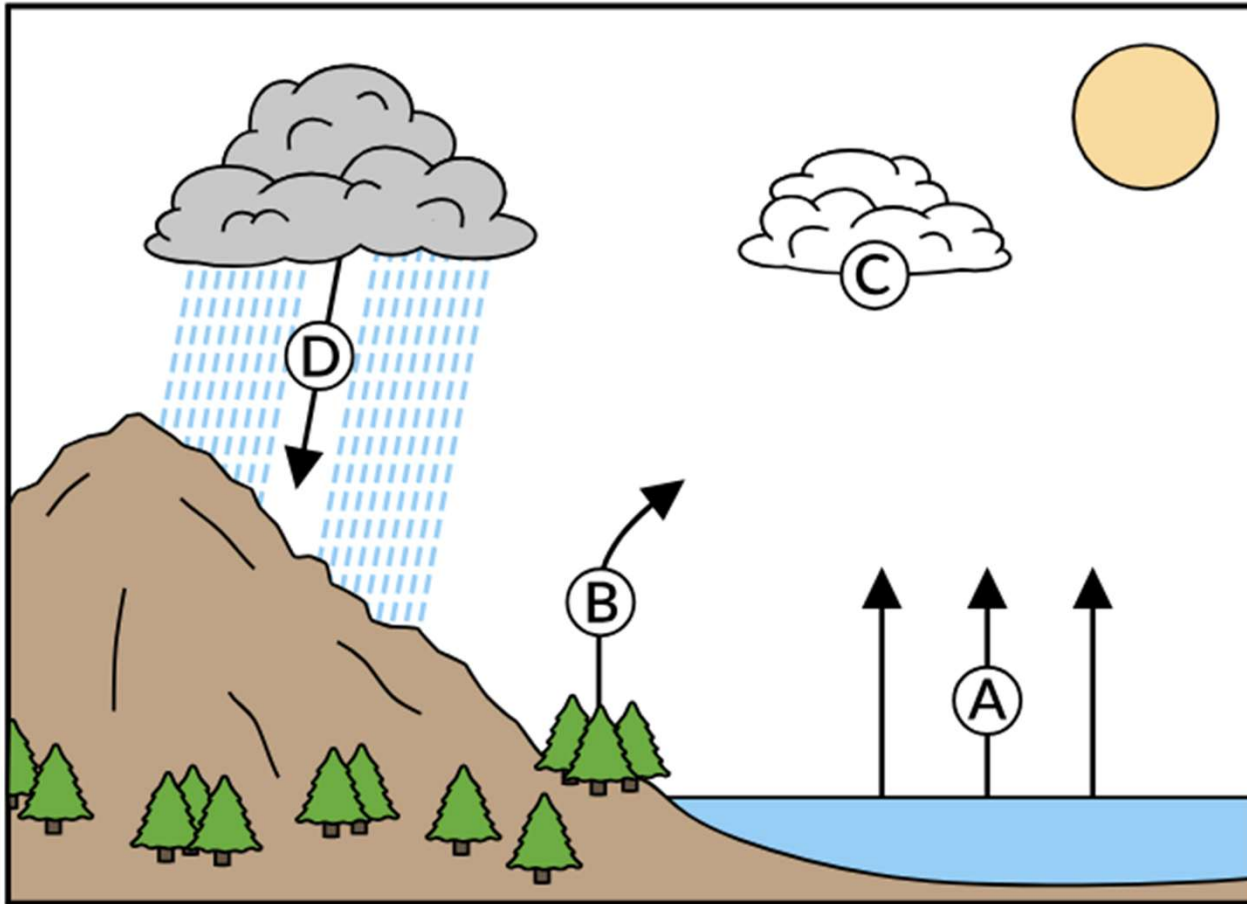
PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 25th August if you want to get them corrected before the Midsem exam

MCQs

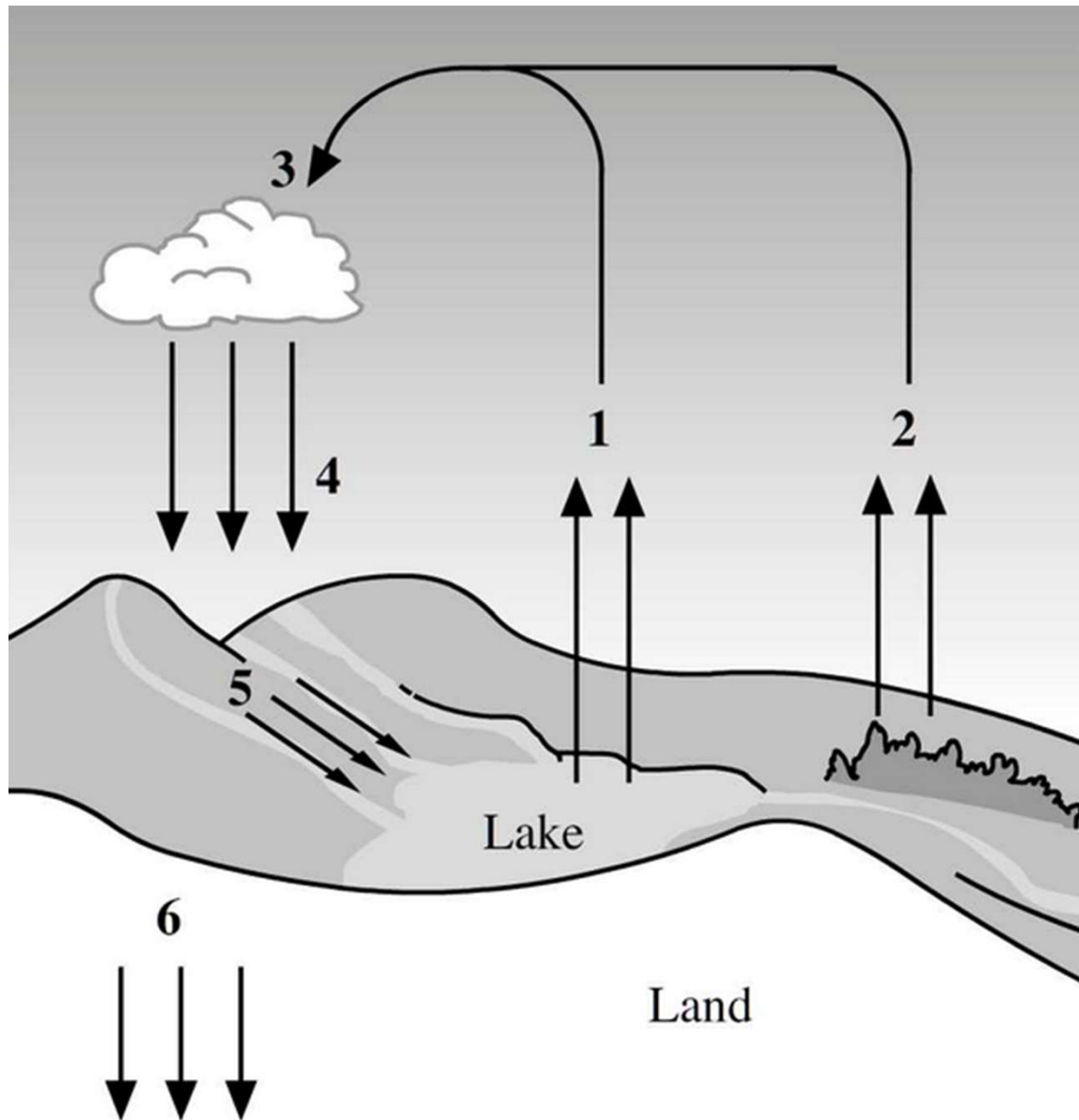
1. Which of the following processes within the water cycle is responsible for changing water from a liquid to a gas?
 - A. Precipitation
 - B. Condensation
 - C. Evaporation
 - D. Groundwater flow
 - E. Surface runoff
2. What percentage of global precipitation falls on the land compared to the oceans?
 - A. 13 percent
 - B. 22 percent
 - C. 35 percent
 - D. 48 percent

Short answer



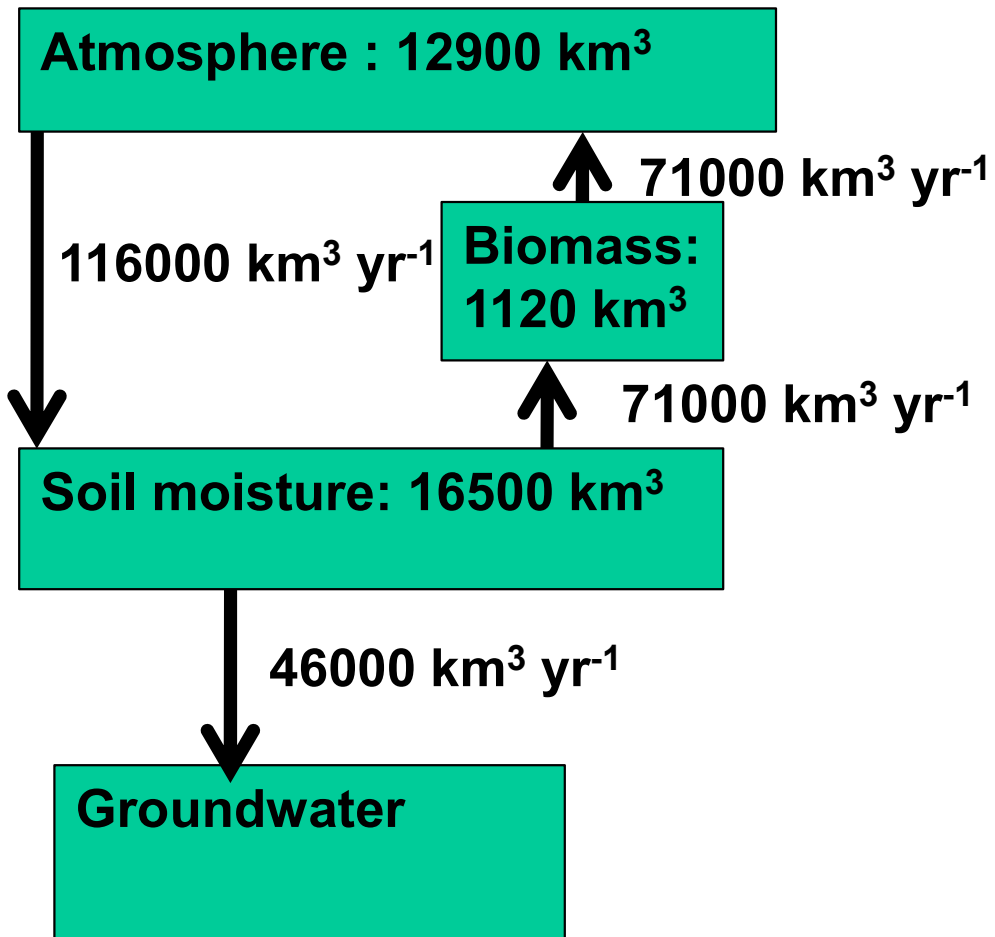
The picture shows a simplified version of the water cycle. Name the processes A, B, C, and D.

Short answer



The picture shows a simplified version of the water cycle. Name the processes 1 to 6.

Numerical



Calculate the turnover time of the soil moisture reservoir in days.
Is the reservoir in equilibrium?